How-To Library

Informatica

Prerequisites for Scanning a Kerberos-Enabled Hive Source

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Abstract

This article explains the prerequisite tasks you must perform before you can use Secure@Source to scan a Kerberos-enabled Hive data store.

Supported Versions

- Secure@Source 4.x

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Overview

Before you scan a Hive data store that is enabled for Kerberos, you must perform the following steps:

1. Copy the Kerberos files.
2. Configure the Data Integration Service properties.
3. Create a cluster configuration object in the Administrator Tool.
4. Create the Hive data store in Secure@Source.

Step 1. Copy the Kerberos Authentication and Configuration Files

Important: The user must have access to Hive and the Hive database.

1. Copy the keytab file to a common directory on the Secure@Source node and all nodes of the external cluster that Secure@Source uses.

   Example: `/opt/keytabs/hiveuser.keytab`
2. Verify that the directory and file name is the same on the Secure@Source node and all nodes of the external cluster that Secure@Source uses.
3. Verify that the keytab file is readable.

   ```
   chmod a+r hive.keytab
   ```
4. Copy the `krb5.conf` file to the following locations:
   - Secure@Source node: `/etc/krb5.conf`
   - If you use the Hadoop Blaze mode, copy to: `<INFA_HOME>/services/shared/security/krb5.conf`
   - On all nodes of the external cluster that Secure@Source uses: `/etc/krb5.conf`

5. Verify that the Key Distribution Center server is known and accessible from all Secure@Source and Secure@Source Hadoop Cluster nodes.

   For example, update `/etc/hosts`.

Step 2. Configure the Data Integration Service Properties

1. If you plan to use the Native mode, obtain a Hive Kerberos ticket using the `kinit` command to log in to the Kerberos authentication system.

   Example:

   ```
   kinit -kt /opt/keytabs/cdh3_hive.keytab hive/cdh3.example.com@EXAMPLE.COM
   ```

2. If you do not plan to use the Native mode, perform one of the following set of steps based on connection type:

   - **JDBC connection**

     1. From the Administrator tool, create a new JDBC Data Profiling Warehouse connection. Use the same user and password credentials as the native Profiling Warehouse Connection.
     2. Select the newly created JDBC connection as the Data Profiling Warehouse database.

   - **Non-JDBC connection**

     1. Install the database client software on the cluster where the Hive source is located.
     2. Modify the `$INFA_HOME/services/shared/hadoop/<hadoop distribution>/infaConf/hadoopEnv.properties` file.
        Example:

        ```
        infapdo.env.entry.oracle_home=ORACLE_HOME=/u01/oracle/product/12.1.0/client_1
        infapdo.env.entry.tns_admin=TNS_ADMIN=$ORACLE_HOME/network/admin
        infapdo.env.entry.ld_library_path=LD_LIBRARY_PATH=.:$HADOOP_NODE_INFA_HOME/
        services/shared/bin:$HADOOP_NODE_INFA_HOME/DataTransformation/bin:
        $HADOOP_NODE_HADOOP_DIST/lib/native:$HADOOP_NODE_INFA_HOME/ODBC7.1/lib:
        $HADOOP_NODE_INFA_HOME/jre/lib/amd64:$HADOOP_NODE_INFA_HOME/jre/lib/amd64/
        server:$HADOOP_NODE_INFA_HOME/java/jre/lib/amd64:$HADOOP_NODE_INFA_HOME/
        java/jre/lib/amd64/server:$HADOOP_COMMON_HOME/lib/native:/u01/oracle/
        product/12.1.0/client_1/lib:$LD_LIBRARY_PATH
        infapdo.env.entry.path=PATH=$HADOOP_NODE_HADOOP_DIST/scripts:
        $HADOOP_NODE_INFA_HOME/services/shared/bin:$HADOOP_NODE_INFA_HOME/jre/bin:
        $HADOOP_NODE_INFA_HOME/java/jre/bin:$HADOOP_NODE_INFA_HOME/ODBC7.1/bin:
        $ORACLE_HOME/bin:$PATH
        ```

3. Configure the following Data Integration Service properties:

   - Hadoop Kerberos Service Principal Name. Example: `hiveuser@example.com`
   - Hadoop Kerberos Keytab. Example: `/opt/keytabs/hiveuser.keytab`
   - Data Integration Service Hadoop Distribution Directory. Enter: `<INFA_HOME>/services/shared/hadoop/<hadoop distribution>`
     Example: `INFA_HOME/services/shared/hadoop/hortonworks_2.6`

4. Recycle the Data Integration Service.

Step 3. Create a Cluster Configuration Object

The cluster configuration enables the Data Integration Service to push mapping logic to the Hadoop environment.

From the Administrator tool, create a cluster configuration object for the Hadoop clusters. During the creation process, specify how you want to import cluster configuration details. You can import in one of the following ways:

- **Import from Archive File**. Use this option when access to the Hadoop distribution managers, such as Ambari or Cloudera Manager, is not available.
- **Import from Cluster**. Use this option when access to the Hadoop distribution managers, such as Ambari or Cloudera Manager, is available.

When you create a Hadoop, Hive, or HDFS data store, you specify the cluster configuration object details.

Creating a Cluster Configuration Object - Import from Archive File

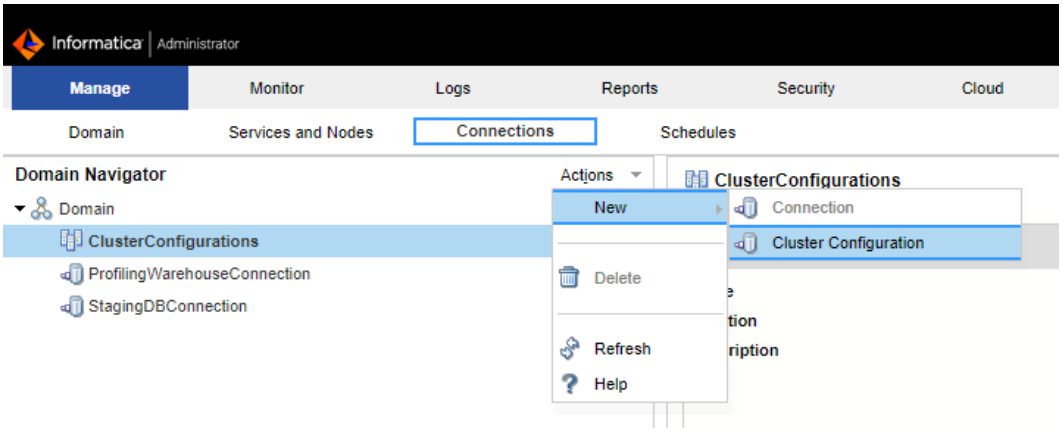
When you create the cluster configuration object, use the **Import from Archive File** option if you do not have access to the Hadoop distribution manager such as Ambari or Cloudera Manager.

Perform the following prerequisite steps:

1. From the Hadoop distribution manager, download the following site-xml files:
   - `core-site.xml`
   - `hive-site.xml`
   - `mapred-site.xml`
   - `tez-site.xml`
   - `yarn-site.xml`
   - `hbase-site.xml`
   - If SSL is enabled: `ssl-client.xml`
2. Compress the site-xml files in a zip file with a clear name such as `HortonworksSiteXMLFiles.zip`
3. Copy the zip file to the machine from which you access the Informatica Administrator tool.

Perform the following steps to create the cluster configuration object:

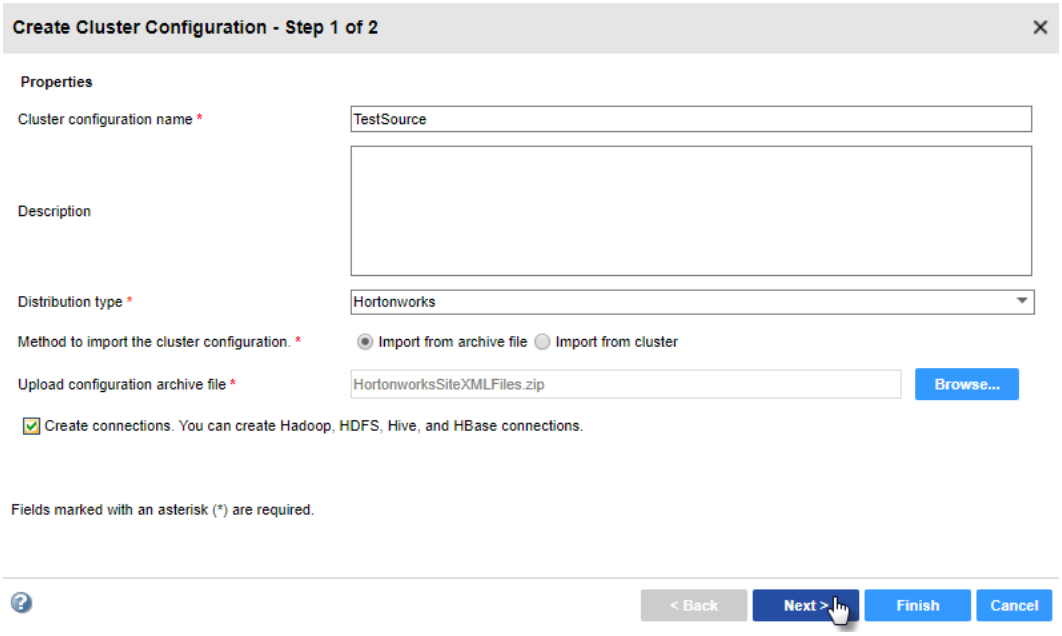
1. Log in to the Informatica Administrator tool.
2. Click the **Connections** tab.
3. In the **Domain Navigator** pane on the left, expand **Domain**.
4. Select Cluster Configurations and click **Actions** > **New** > **Cluster Configuration**.



The **Create Cluster Configuration - Step 1 of 2** window appears.

5. Enter a name for the cluster configuration.

6. Optionally, enter a description.
7. From the drop-down list, select the distribution type.
8. Enable the **Import from archive file** option.
9. Click **Browse** in the **Upload configuration archive file** field and select the zip file with the site-xml files.
10. Check the **Create Connections** option.
11. Click **Next**.



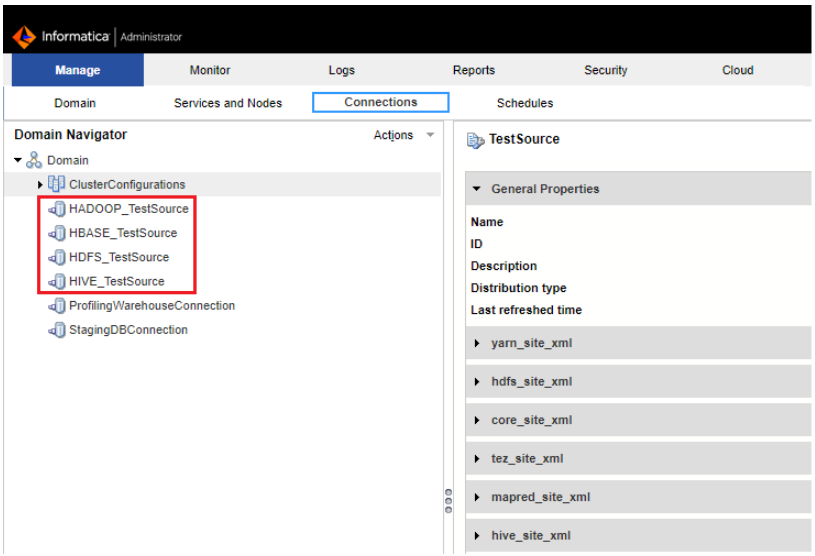
The **Create Cluster Configuration Summary Page - Step 2 of 2** window appears with the details you entered.

12. Review the information and click **Finish**.

    The Hadoop, HDFS, Hive, and HBase connections are created with the following name format:

    - `HADOOP_<Cluster Configuration Name>`
    - `HBASE_<Cluster Configuration Name>`
    - `HDFS_<Cluster Configuration Name>`
    - `HIVE_<Cluster Configuration Name>`

For example if the cluster configuration name is `TestSource`, the connection names would appear as in the following image:



Creating a Cluster Configuration Object - Import from Cluster

When you create the cluster configuration object, use the **Import from Cluster** option if you have access to the Hadoop distribution manager such as Ambari or Cloudera Manager.

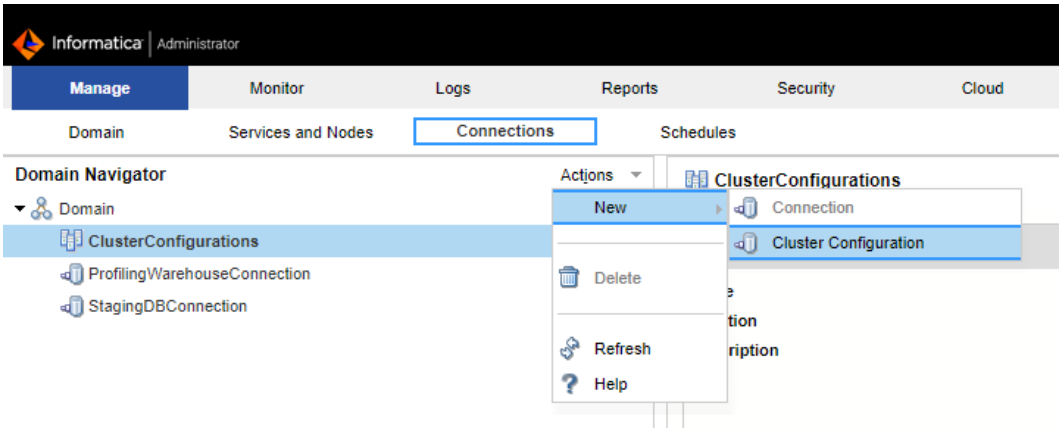
Before you create the cluster configuration object, note the following details about the Hadoop cluster:

- Administrator host name.
- Administrator port number.
- Administrator user ID.
- Administrator password.
- Hadoop cluster name.

Perform the following steps to create the cluster configuration object:

1. Log in to the Informatica Administrator tool.
2. Click the **Connections** tab.
3. In the **Domain Navigator** pane on the left, expand **Domain**.

4. Select Cluster Configurations and click **Actions** > **New** > **Cluster Configuration**.

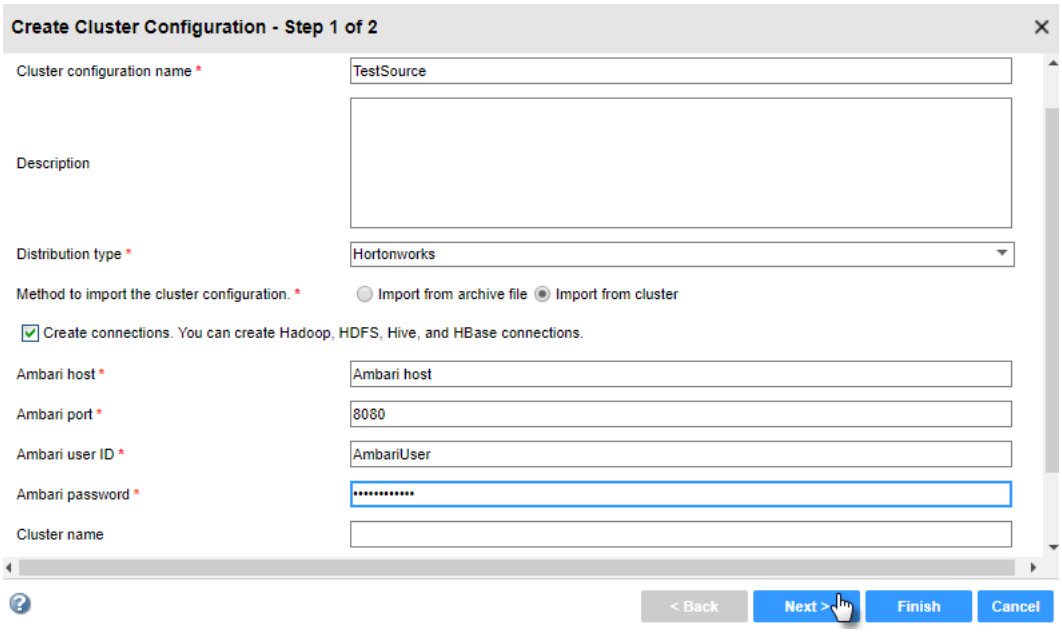


The **Create Cluster Configuration - Step 1 of 2** window appears.

5. Enter a name for the cluster configuration.
6. Optionally, enter a description.
7. From the drop-down list, select the distribution type.
8. Enable the **Import from cluster** option.
9. Check the **Create Connections** option.
10. To import from the cluster, enter values for the following properties for Ambari or Cloudera Manager:

Name	Description
Host	The hostname of the machine on which Ambari or Cloudera Manager is running.
Port	The port on which Ambari or Cloudera Manager is running. Default for Ambari is 8080. Default for Cloudera Manager is 7180.
User ID	The user name for logging into Ambari or Cloudera Manager.
Password	The password for the Ambari or Cloudera Manager user.
Cluster Name	Optional field to provide the cluster name in Ambari or Cloudera Manager. Do not provide the value if Ambari or Cloudera Manager is not configured for multiple clusters. If you still want to provide a value, get the exact cluster name from your administrator.

11. Click **Next**.



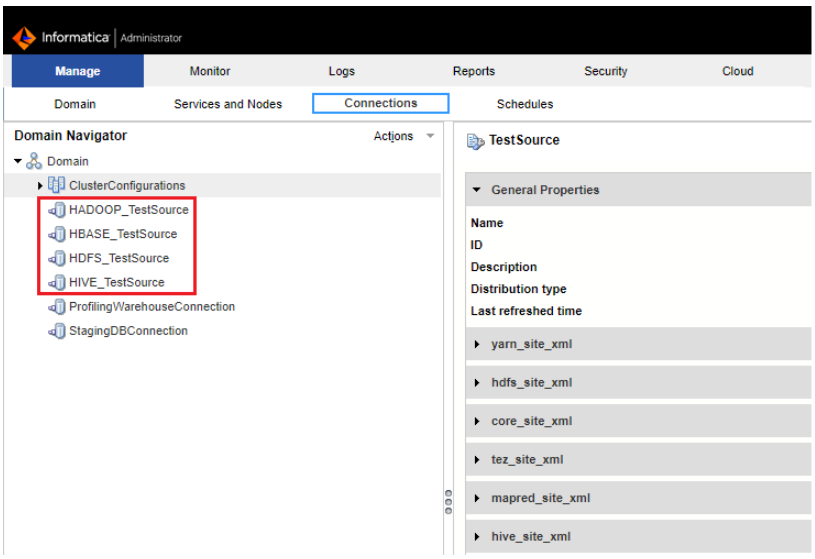
The **Create Cluster Configuration Summary Page - Step 2 of 2** window appears with the details you entered.

12. Review the information and click **Finish**.

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- `HADOOP_<Cluster Configuration Name>`
- `HBASE_<Cluster Configuration Name>`
- `HDFS_<Cluster Configuration Name>`
- `HIVE_<Cluster Configuration Name>`

For example if the cluster configuration name is `TestSource`, the connection names would appear as in the following image:



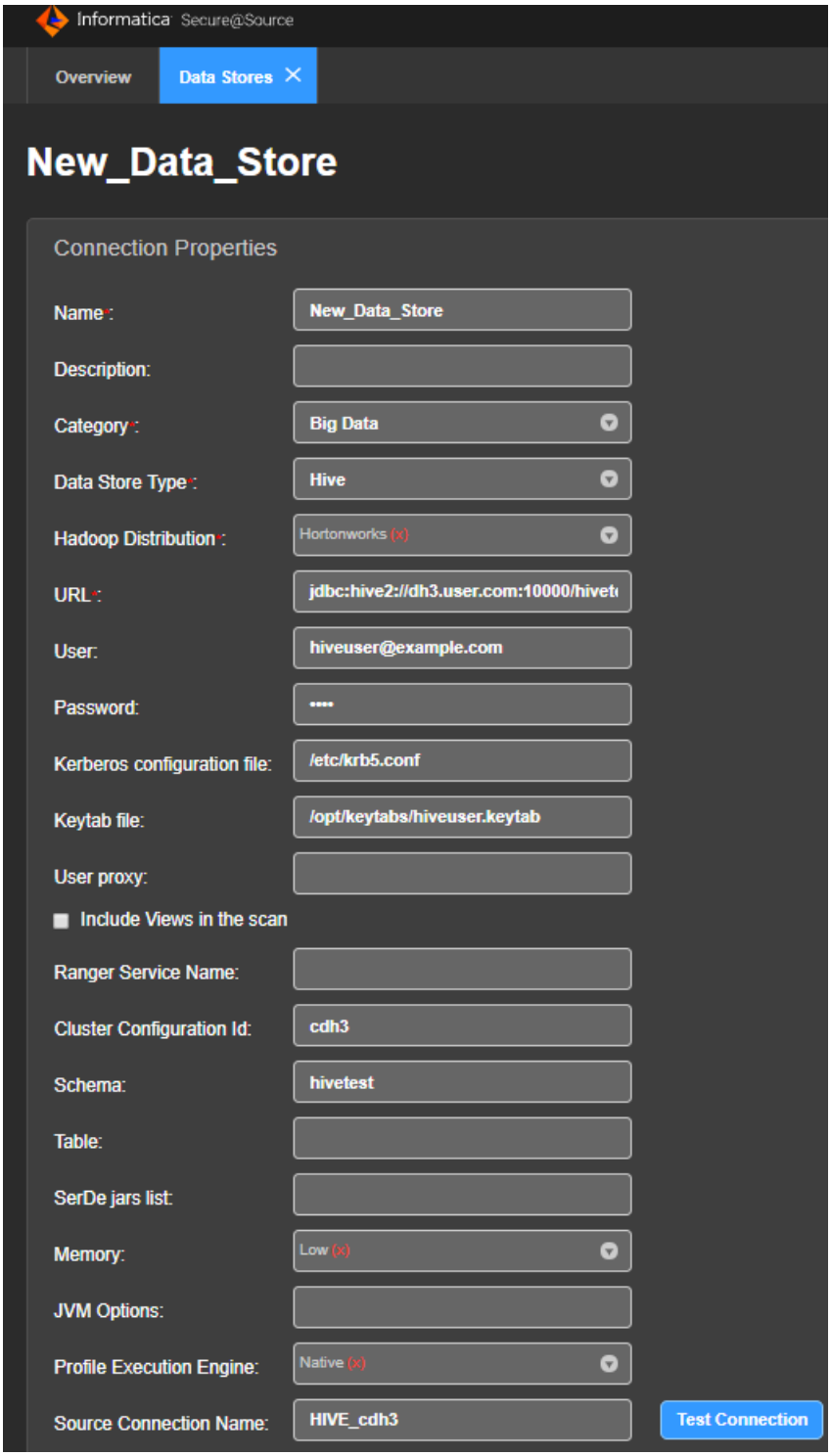
Step 4. Create a Hive Data Store

1. From Secure@Source, click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
2. Click **New**.

   The **New Data Store** page appears.
3. Configure the properties.

The following image shows a sample data store page:



4. Click **Save**.

   You can profile the data store by adding it to a scan.

Author